

Question_2:

Question description

saran, subash, and Yasir alias Pari are three first-year engineering students of the State Technical Institution (STI), India. While saran and subash are average students who come from a Middle class, Yasir is from a rich family. saran studies, engineering as per his father's wishes, while subash, whose family is poor, studies engineering to improve his family's financial situation.

Yasir, however, studies engineering of his simple passion for developing android applications.

Yasir is participating in a hackathon for android application development. the task is Insertion in a Doubly Linked list at beginig.

Input Format

First line indicates the number of elements N to be inserted in array

Second line indicates the array elements according to the N

Output Format

First line represents the doubly linked list in forward direction

Second Line represents the doubly linked list in backward direction

Logical Test Cases

Test Case 1

INPUT (STDIN)

16
1 3 4 7 9 11 13 14 18 19 23 24 56 32 98 17

EXPECTED OUTPUT

17 98 32 56 24 23 19 18 14 13 11 9 7 4 3 1
1 3 4 7 9 11 13 14 18 19 23 24 56 32 98 17

Test Case 2

INPUT (STDIN)

15
0 1 2 0 1 1 1 0 2 0 2 2 0 1 2

EXPECTED OUTPUT

2 1 0 2 2 0 2 0 1 1 1 0 2 1 0
0 1 2 0 1 1 1 0 2 0 2 2 0 1 2

Code:

```
1  /*
2  QUESTION_2:
3  Insert a new node at the beginning of a Doubly Linked List.
4  */
5  #include <iostream>
6  using namespace std;
7
8  struct Node {
9      int data;
10     Node *next;
11     Node *prev;
12 };
13
14 void insertStart(Node *&head, int x) {
15     Node *p = new Node{x, head, NULL};
16     if (head)
17         head->prev = p;
18     head = p;
19 }
20 int main() {
21     int n, x;
22     cin >> n;
23
24     Node *head = NULL;
25     for (int i = 0; i < n; i++) {
26         cin >> x;
27         insertStart(head, x);
28     }
29     Node *p = head;
30
31     while (p) {
32         cout << p->data << " ";
33         p = p->next;
34     }
35     cout << endl;
36     p = head;
37
38     while (p->next)
```

```
39         p = p->next;
40     while (p) {
41         cout << p->data << " ";
42         p = p->prev;
43     }
44     return 0;
45 }
```

Output:

Test Case_1:

```
16
1 3 4 7 9 11 13 14 18 19 23 24 56 32 98 17
17 98 32 56 24 23 19 18 14 13 11 9 7 4 3 1
1 3 4 7 9 11 13 14 18 19 23 24 56 32 98 17
PS C:\Users\Prabin Yadav\Documents\III SEM\D
```

Test Case_2:

```
15
0 1 2 0 1 1 1 0 2 0 2 2 0 1 2
2 1 0 2 2 0 2 0 1 1 1 0 2 1 0
0 1 2 0 1 1 1 0 2 0 2 2 0 1 2
PS C:\Users\Prabin Yadav\Document
```